SIGHT Project - Meeting 28

Project Name: Safety Immersion and Gamified Hazard Training for Industry 5.0 Workers

Project Number: WSIC26-250206-009

Date: October 30, 2026

Time: 11:00 AM to 12:03 PM EDT

Location: Zoom

Meeting Focus: Recap of Meeting 27 action items, one participant ID across VR and AR, the Q5 invoice and the IRB amendment, BWC deliverables and the Track 3 abstract, team requirements, the experimental plan and its assessments, session time, compensation and retention, recruitment and voluntary participation, and task ownership.

Attendees

- **Miami University (MU):** Fadel Megahed (PI, Meeting Chair), Arthur Carvalho (Co-PI), Mohammad Mayyas (Co-PI), Jay Shan (Co-PI), Ibrahim Yousif (Associate Research Scientist), Michael Wise (GRA), Austin Hamilton (GRA/SIGHT AR Lead)
- **Discovery Center:** Kristen Morio (Evaluator), Abby Helsinger (Senior Research Associate/Evaluator), Yue Li (Evaluator)
- **External Consultant:** Lora Cavuoto (University at Buffalo)

1. Introductory Remarks and Recap / Approval of Minutes

- **Welcome & Recap:** Fadel Megahed chaired the meeting. Fadel thanked Mohammad Mayyas for chairing Meeting 27. Fadel summarized the four project technologies: the AI backbone, the web-based VR, the instructional AR for the three machines, and the digital twin that verifies machine actions. Fadel stated that the project is in its final stretch and that the focus now moves to the experimental plan.
- **Approval of Minutes:** The Meeting 27 minutes and the Meeting 28 agenda were available on Google Drive. The team raised no corrections.
- **Status of Meeting 27 Action Items:**
    - AR evaluation data with the VR backend and one participant ID: open. Arthur Carvalho explained that each VR user gets an ID from Firebase. The AR has no sign-in at this time.
    - Arthur proposed that the AR use the same sign-in as the VR (Google or email and

password), with a separate AR database. Austin Hamilton asked for two databases that share the user ID. Arthur agreed and stated that the VR production database must not change.

- Arthur showed the Firebase settings that the AR needs. Austin must give Arthur the AR web domain before deployment, so that Arthur can add it to the authorized domains. Arthur noted that Google Analytics will then combine VR and AR use.
- Fadel asked for a sign-in option for participants who do not have a Google account. Participants do the VR session first, so the researcher can give each participant the same login for the AR session.
- Ibrahim Yousif asked why the team links the two sessions. Fadel explained that the link shows that the same person completed both sessions, which counts toward the 36 participants, and it supports analysis across VR and AR. Lora Cavuoto added that participants answer the baseline questions only once.
- Tests of the vision model in extreme conditions: done. Ibrahim reported that the team tested more people and different clothing in response to a paper reviewer comment.
- Contact with Jeff Malek about cost-benefit data: not confirmed. Leo Lyublinski did not attend.
- Mill SOP updates: done. Jay Shan volunteered to test the mill SOP.
- Protocol meeting with Lora, Jay, and Kristen Morio: held on October 23.
- CITI training for Usman Asif and Om Raval: open. Mohammad had no reply from them yet.

2. Administrative Updates

- **Q5 Expenditure Report and Invoice:** The report and the invoice were due on October 30. Paula Murray planned to submit them by noon. Fadel answered her one question.
- **IRB Amendment:** The amendment must add the new hires, and each new hire needs CITI training. Miami moved IRB submissions to a new online system, and Lora does not have access to it. Fadel asked Jay to coordinate the amendment and submit it by the end of the following week.
- **BWC Updates:**
  - Fadel, Ibrahim, and Arthur attended the quarterly grant meeting on October 22. SIGHT presented first, and BWC received the presentation well.
  - BWC stated that the project presentation video (Deliverable 9.2) takes time. BWC shared an example video. Fadel will send the example to the team.

- The two-page Track 3 abstract is due in 17 days (November 16, 5 PM ET).
- BWC approved the Q5 report and the grant meeting deliverable. The September/October meeting documents stay pending until all teams submit.

- **Team Requirements:**
  - Paula Murray informed Fadel that each person on the grant, including undergraduate assistants, must complete the COI training and the FCOI form. The links are in the deck, and Fadel will also send them by email.
  - The deadline for CITI training for the new hires is November 1. The deadline for their machine training is November 4. Team members cannot run the experiment without CITI training.
  - Team members must file each receipt in the PCard folder on Google Drive, by month and by vendor, when Fadel or Jacob asks for it.
  - Team members must write emails in their own voice. Fadel stated that AI can polish text, but fully AI-generated emails are not acceptable.
- **Makino Update:** Mohammad contacted David Fox and the Makino team about a visit. The purpose of the visit is to present the VR modules that Arthur developed and to discuss commercialization pathways. Makino gave informal feedback and showed interest. Makino has not confirmed a date. Mohammad sent a follow-up on October 30 and asked the Miami workforce liaison to lead the formal relationship.

3. Technical Updates: Experimental Plan

- **How We Got Here:**
  - Fadel listed the meetings that discussed the experiment: July 24, the September 11 breakout, October 2, October 16, and the October 23 protocol meeting. Fadel asked the team to follow up faster on decisions.
  - Fadel thanked Ibrahim for a detailed comparison of the IRB documents with the proposal.
  - Fadel will add short video clips from earlier meetings to the decks, to remind the team of agreed decisions. One clip showed the September 11 decision: participants train in VR first and then in AR, which reduces risk.
  - Ibrahim stated that an earlier decision on operational AR could have saved the weeks that Michael Wise spent on machine guards. Fadel explained that the team will build operational AR to show readiness for the next grant. The human subjects experiment

does not have to assess it.

- **Assessments:**
  - Before the session: informed consent, demographics, the worker safety commitment items of the Nordic questionnaire (reworded for participants), possible safety knowledge items, the OSHA Job Hazard Analysis quiz, and the GATT questionnaire. The baseline questions occur only in Session 1.
  - Fadel proposed that the researcher repeats the participant rights at the start of Session 2.
  - During the session: the VR assessment modules and an observation sheet. The observation sheet records clock times, so that the team can match notes to the system logs. The AR assessments are not yet agreed.
  - After the session: the TAM questionnaire and the interview questions. Audio recording of the interview needs IRB approval.
  - The IRB amendment must also change the session order, replace the HoloLens with the Magic Leap 2, and correct the email domain in the consent form. Ibrahim found the domain error.
  - Kristen stated that the evaluation team must be on the IRB to use the data for its secondary analysis. Kristen asked the team to add Abby Helsinger and Yue Li for now. Kristen will review the IRB first and can join it later.
  - The IRB asked for non-operational machines in AR. Fadel stated that an assessment of one module per machine can use non-operational machines, but this reduces the work that participants do with the headset. The team may meet with the IRB about this.
- **Time Assessment:**
  - Fadel proposed that the amendment states a session of about two hours, in place of 90 minutes.
  - Ibrahim reported, from the times that Michael and Austin recorded, that most modules take less than two minutes and the longest VR modules take four to six minutes. Ibrahim considers the AR the only time risk.
  - Lora confirmed that participants review all modules and complete one assessment module per machine. Lora stated that the session length is an estimate, and a shorter session is acceptable.
- **Compensation and Retention:**
  - Fadel confirmed that participants receive $75 for each session.

- Ibrahim asked how the team can make sure that 90% of participants complete both sessions. Fadel explained that informed consent does not allow the team to hold participants. The team will recruit until 36 participants complete both sessions.
- Lora stated that a 10% drop-off is typical. This is why the team recruits 40 participants for 36 completions.

- **Recruitment and Voluntary Participation:**
  - Fadel used a live game with three questions about an email from the ENT 252 instructor. The email proposed that participation be part of the course final. Some participants could not select answers on their phones.
  - Fadel stated that participation cannot be part of a final or a grade, because participation must be voluntary. The instructor can tell students about the study and the compensation, and can encourage them to volunteer.
  - Arthur added two concerns: the VR is not yet fully accessible to screen readers, and as the database administrator Arthur could see the grades, which is likely a FERPA problem.
- **Task Ownership:**
  - Ibrahim asked the team to assign one owner to each task, who consults the other experts.
  - Ibrahim reported that the AM Hub lab monitors are 23.7 to 24 inches, but the IRB states 27-inch monitors. The team must change the IRB or get monitors. This also reduces the VR session capacity.
  - Fadel agreed that the team needs clear owners. Jay leads the IRB amendment, and Jay needs the CITI certificates of Usman and Om for it.
  - Ibrahim did not have time to present.

4. Industry Partner Collaboration

- **MaxByte:** Ramshankar C S sent an email that he could not attend. Fadel replied before the meeting with two requests. Mohammad will follow up on the two items.
- **Industrious Group Inc.:** Jeff Malek did not attend. The team will confirm whether Leo contacted Jeff about the preliminary data for the cost-benefit model.

5. Risk Mitigation and Log Updates

- **IRB Amendment:** This risk remains open. The amendment must cover the new hires, the session order, the headset, the compensation, the session length, and the monitor size before testing starts.
- **Voluntary Participation (new):** Recruitment through a course must not connect participation to a grade. The team will use only the recruitment methods that the IRB approved.
- **Monitor Size (new):** The lab monitors do not match the 27-inch monitors in the IRB. The team must change the IRB or get monitors.
- **Participant Retention:** The team expects a 10% drop-off and will recruit until 36 participants complete both sessions.
- **Task Ownership (new):** Tasks without a clear owner slow the experiment. The team will name an owner for each task.

6. Further Discussion and Wrap-Up

- **Next Meeting:** The next bi-monthly meeting is on November 13, 2026. This meeting is reported to BWC. Fadel will send the agenda early and asked for technical updates. Fadel wants pre-pilot testing to start by then.
- **Breakout:** From about 12:08 to 12:45 PM, Fadel, Arthur, Ibrahim, Jay, and Lora met to discuss decisions on the testing protocol. The breakout was not transcribed.

Action Items

Task	Responsible	Deadline
Coordinate and submit the IRB amendment	Jay Shan	November 6, 2026
Complete CITI training and send the certificates to Jay	Usman Asif / Om Raval	November 1, 2026
Complete the COI training and the FCOI form	All team members	ASAP
Connect the AR sign-in to Firebase and use one participant ID across VR and AR	Austin Hamilton / Arthur Carvalho	Before pre-pilots

Task	Responsible	Deadline
Agree on the AR assessments and whether AR machines operate	Fadel Megahed / Arthur Carvalho / Lora Cavuoto	Before pre-pilots
Resolve the monitor size (IRB change or new monitors)	Jay Shan	With the amendment
Reply to the ENT 252 instructor: participation must be voluntary and not graded	Mohammad Mayyas	November 2, 2026
Submit the Track 3 abstract	Fadel Megahed	November 16, 2026 (5 PM ET)
Follow up on the MaxByte items and the Makino visit	Mohammad Mayyas	ASAP
Start pre-pilot testing	MU team	November 13, 2026